

ZHOU ZHOU

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SUMMARY

Master's candidate in **Data Analytics** at George Mason University with applied experience in AI, machine learning, and predictive analytics. Proficient in **Python, SQL, NLP, and BI tools**, delivering actionable insights that optimize financial performance, operations, and decision-making across finance, real estate, and technology.

WORK & LEADERSHIP EXPERIENCE

Willow Bridge

Reston, VA

Data Analyst – Finance, budgeting and accounting

03/22- Present

- Utilized **Python and PySpark** to process 1M+ construction project, budget, and cost-variance records, resolving missing cost and schedule fields via K-Means-based imputation and forward-fill logic; standardized multi-source date formats and applied **Isolation Forest** to flag abnormal cost overruns, improving forecast reliability and risk visibility.
- Queried and analyzed large-scale budget, invoice, and vendor datasets using **BigQuery and SQL**, extracting features such as cost burn rate, change-order frequency, and labor-material mix; applied **trend and variance analysis** to inform budget reallocation strategies, while optimizing joins, CTEs, and partition pruning to cut daily refresh time by **90%**.
- Developed machine-learning models (**XGBoost, Random Forest**) to predict project cost overruns and schedule delays, incorporating historical spend patterns, vendor performance, and project attributes; generated **early-warning signals that guided mitigation strategies**, reducing unexpected budget deviations and **improving capital planning accuracy**.
- Built **Power BI dashboards** to monitor project KPIs including budget utilization, forecast-to-actual variance, and ROI by project type, conducted cohort and trend analysis to identify systemic inefficiencies, and enhanced **DAX calculations and drill-down filters** by region and contractor, improving executive reporting efficiency by **70%**.
- Automated personalized executive and project-manager reporting using **Python and Databricks**, generating and distributing Excel reports with insights on budget risk exposure, forecast accuracy, and vendor performance; built reusable reporting and metrics functions to ensure timely delivery and support faster financial and investment decisions.

Gates Hudson

Tysons, VA

Portfolio Data Analyst

08/20 – 03/22

- Built predictive cash-flow forecasting models using **Python (ARIMA, rolling regression)** on multi-year transaction and rent-collection data, improving liquidity planning and reserve allocation; identified seasonality and tenant payment patterns that reduced cash shortfalls by 5% across the portfolio.
- Queried and analyzed mortgage origination and performance data using **SQL**, extracting features such as LTV, DSCR, delinquency duration, and interest-rate sensitivity; identified high-risk borrower segments and underwriting gaps, generating insights that **improved approval screening and reduced early-stage default exposure**.
- Developed insurance payment and claims analytics using **Logistic Regression**, modeling late-payment risk and claim escalation probability; identified vendors and property types driving excess claims, informing renegotiation and coverage optimization strategies that lowered penalty risk and improved procurement decisions.
- Conducted **real estate tax optimization analysis** by benchmarking assessed values against market comps and historical reassessments, building a data-driven evaluation framework to prioritize appeals and compliance checks, delivering sustained **annual property-tax savings exceeding 15%** across managed assets.
- Designed **Tableau dashboards** monitoring portfolio KPIs, including cash coverage ratio, delinquency rates, mortgage performance trends, tax variance, insurance claim frequency, and payment cycle time; enhanced stakeholder engagement by translating complex financial signals into **actionable asset-management insights**.
- Implemented **A/B testing** to evaluate alternative cash-reserve and payment-timing strategies, defining success metrics such as liquidity coverage, late-fee incidence, and volatility reduction; applied **t-tests and confidence intervals** to validate improvements, guiding **adoption of optimized financial controls across properties**.

EDUCATION

George Mason University

Fairfax, VA

Master of Science in Data Analytics Engineering (Dean's List)

08/24 - present

- GPA:** [3.8] / 4.0

SKILLS, ACTIVITIES & INTERESTS

- Marketing Related:** A/B Test, Google Analytics, CRM, Conversion Lift Testing, Marketing Mix Modeling
- Analysis & Visualization:** Python (Pandas, NumPy, Scikit-Learn), R, Tableau, Power BI, Looker Studio, Excel
- Database:** Snowflake, Big Query, AWS (EC2, Glue, S3), Oracle SQL, Microsoft Azure, MySQL, Apache Airflow
- Statistics & Machine Learning:** Hypothesis Testing, Naive Bayes, Causal Inference, Linear Regression, Logistic Regression, NLP, Sentiment Analysis, Time Series, Boosting Algorithms, Recommendation System, SVM, Cluster

PROJECT

AI-Powered Resume Optimizer for LinkedIn Jobs

- Built semantic resume-job matching using Sentence Transformers, quantifying fit via similarity scores, metrics dashboards.
- Engineered automated keyword extraction and skill relevance scoring to evaluate predictive job-fit accuracy performance.
- Deployed Streamlit app integrating GPT generation for tailored cover letters, improving alignment by 30%.